

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

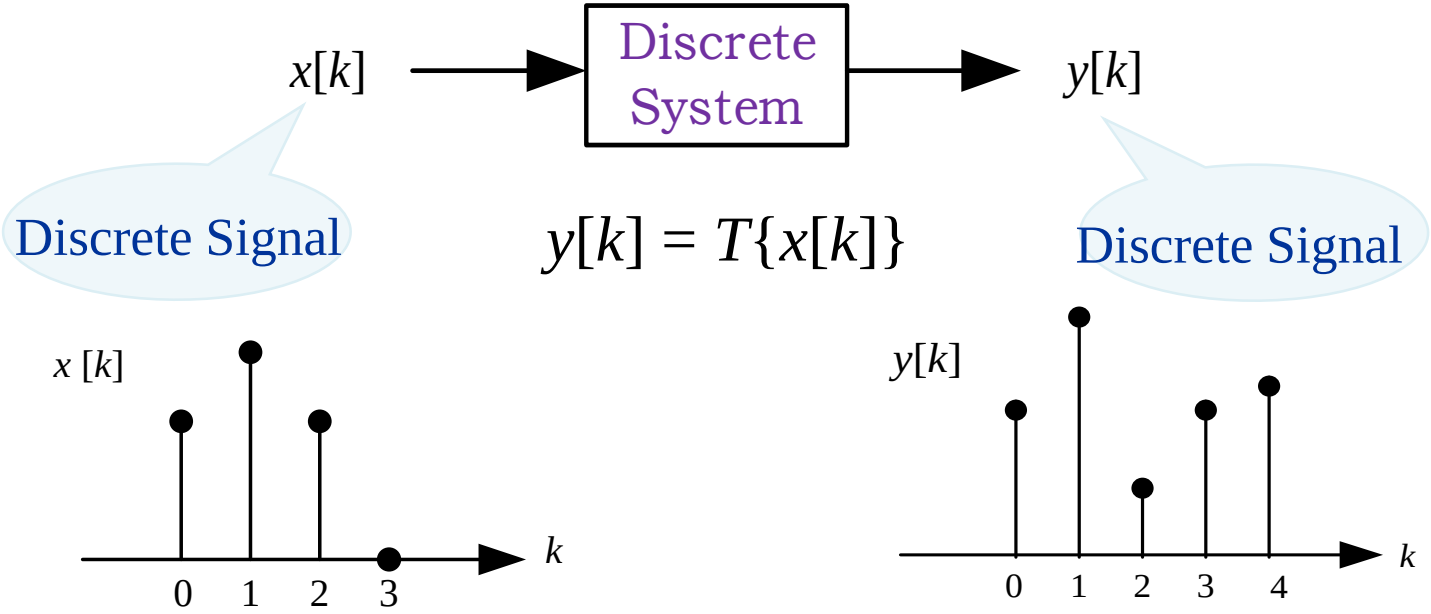
Time domain analysis of discrete system



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- Definition and Classification of Discrete Systems
- Time-Domain Description of Discrete Linear Time-Invariant (LTI) Systems
- Analysis of Responses in Discrete LTI Systems
- Engineering Applications of Sequential Convolution Sums

Definition of Discrete System:





- Linear and nonlinear systems
- Time varying system and time-invariant systems
- Causal and non-causal systems
- Stable and unstable Systems

- Linear system: a system with linear characteristics。

Uniform characteristics and superposition characteristics

Only when a system has both uniform characteristics and superposition characteristics can it be called a linear system, which can be expressed as：

$$T\{ax_1[k]+bx_2[k]\}=aT\{x_1[k]\}+bT\{x_2[k]\}$$

- Nonlinear system: a system that does not have linear characteristics.

- Time invariant system: The relationship between the zero-state response of the system and the input excitation does not change with the starting point at which the input excitation acts on the system.

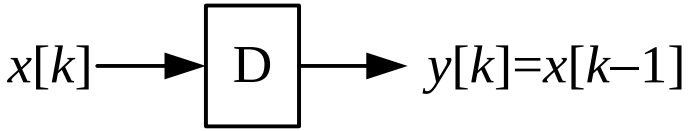
A discrete time-varying system is represented as：

$$\text{If} \quad x[k] \longrightarrow y_{zs}[k] \quad \text{Then} \quad x[k-n] \longrightarrow y_{zs}[k-n]$$

- Time-varying system: A system that does not satisfy the property of time-invariance.

[E.g] Try to Please determine whether *a* unit delay element is a linear time-invariant system.

Solution: (1) Linearity Characteristic :



$$x_1[k] \Rightarrow y_1[k] = x_1[k-1]$$

$$x_2[k] \Rightarrow y_2[k] = x_2[k-1]$$

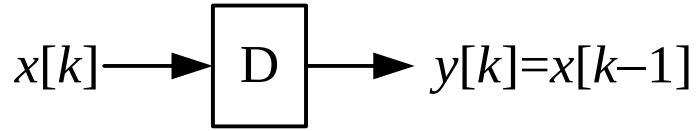
$$x[k] = \alpha x_1[k] + \beta x_2[k]$$

$$y[k] = \alpha x_1[k-1] + \beta x_2[k-1] = \alpha y_1[k] + \beta y_2[k]$$

Since the system possesses linear characteristics, the unit delay element is a linear system.

[E.g] Try to lease determine whether the unit delay element is a linear time-invariant system.

Solution: (2) Time-Invariance Characteristic



$$x[k] \quad \Rightarrow \quad y[k] = x[k-1]$$

$$x_1[k] = x[k-k_0]$$

$$y_1[k] = x_1[k-1] \; = x[k-k_0-1] \; = y[k-k_0]$$

The unit delay element is a time-invariant system.

Therefore, the unit delay element is a linear time-invariant system.

[E.g] Try to determine whether the unit delay element is a linear time-invariant system

Solution: From this, we can deduce that the input-output relationship of the system is given by:

$$y[k] = a_0 x[k] + a_1 x[k-1] + \cdots + a_M x[k-M]$$

Therefore, it can be inferred that the system with the unit delay element.

For example, the input-output relationship of the system can be represented as:

$$y[k] = 0.2x[k+2] + 0.2x[k+1] + 0.2x[k] + 0.2x[k-1] + 0.2x[k-2]$$

The system represented by the unit delay element.

[E.g] Try to determine whether a decimator is a linear time-invariant (LTI) system

Solution: Taking a decimation factor of 2 as an example, the input-output relationship can be described as:

$$y[k] = x[2k]$$

(1) Linear characteristic :

$$x_1[k] \Rightarrow y_1[k] = x_1[2k]$$

$$x_2[k] \Rightarrow y_2[k] = x_2[2k]$$

The system possesses linear characteristics, therefore, the decimator is a linear system.

$$x[k] = \alpha x_1[k] + \beta x_2[k]$$



$$y[k] = \alpha x_1[2k] + \beta x_2[2k] = \alpha y_1[k] + \beta y_2[k]$$

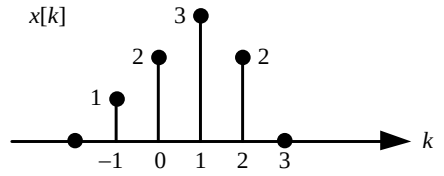
[E.g] Try to determine whether the decimator is a linear time-invariant system?

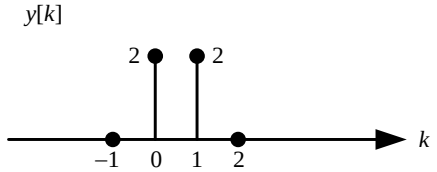
Solution: Taking a decimation factor of 2 as an example, the input-output relationship can be described as:

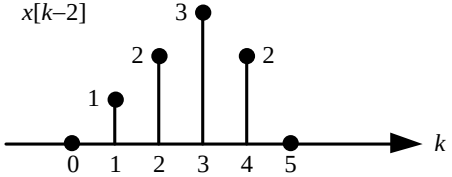
$$y[k] = x[2k]$$

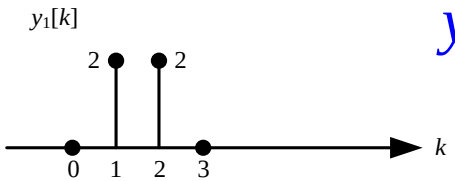
(2) Time-Invariance Characteristic Evaluation

The decimator is a time-varying system.



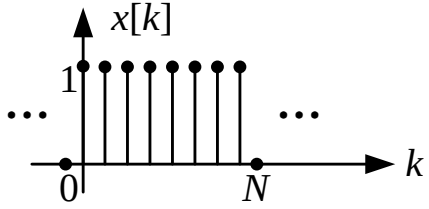


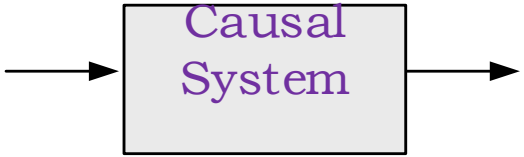




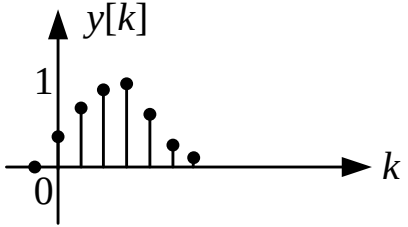
$$y_1[k] \neq y[k-2]$$

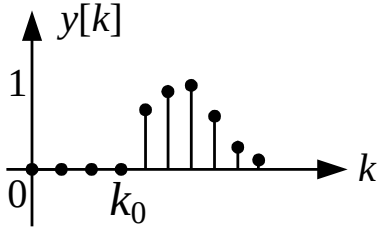
- Causal System: The output response of the system does not precede the input signal of the system.。





Causal System Diagram



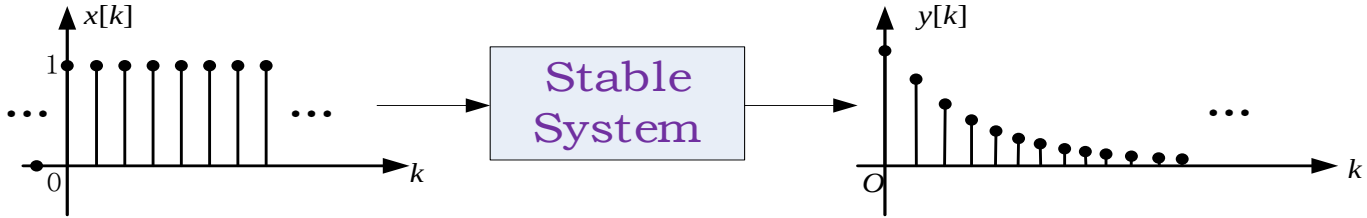


- Non-causal System: A system that does not possess causal properties is referred to as a non-causal system.

➢ Stable System:

A system that produces bounded outputs for any bounded inputs is known as a Bounded-Input Bounded-Output (BIBO) stable system.

BIBO: Bounded Input, Bounded Output



➢ Unstable System: When the input to a system is bounded, the output may be unbounded.

Mathematical Model:

The input-output relationship of a discrete Linear Time-Invariant (LTI) system can be described by a linear constant-coefficient difference equation.

$$a_0 y[k] + a_1 y[k-1] + \cdots + a_{N-1} y[k-N+1] + a_N y[k-N]$$
$$= b_0 x[k] + b_1 x[k-1] + \cdots + b_{M-1} x[k-M+1] + b_M x[k-M]$$

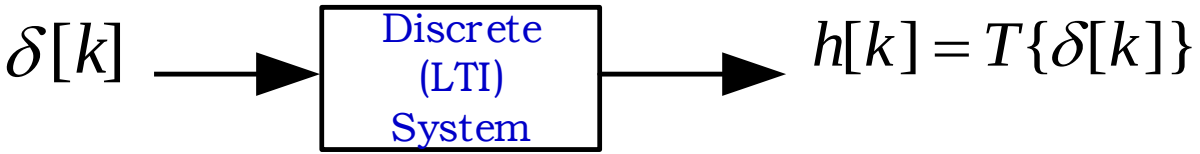
or
$$\sum_{i=0}^{N} a_i y[k-i] = \sum_{j=0}^{M} b_j x[k-j]$$

which: $a_i (i = 0, 1, \cdots, N), b_j (j = 0, 1, \cdots, M)$

Time-Domain Description:

The unit impulse response $h[k]$ is the time-domain description of a discrete Linear Time-Invariant (LTI) system.

$h[k]$ is the zero-state response of the system when excited by a unit impulse sequence $\delta[k]$.



(Zero initial state)

A necessary and sufficient condition for a discrete Linear Time-Invariant (LTI) system to be causal is that its unit impulse response h[k] is:

$$h[k] = 0, k < 0$$

A necessary and sufficient condition for a discrete LTI system to be stable is that:

$$\sum_{k=-\infty}^{\infty} \left| h[k] \right| = S < \infty$$

[E.g] Assess the causality and stability of the M_1+M_2+1 point moving average system.

Solution: The input-output relationship of the M_1+M_2+1point moving average system is given by :

$$y[k]=\frac{1}{M_1+M_2+1}\sum_{n=-M_1}^{M_2}x[k+n]$$

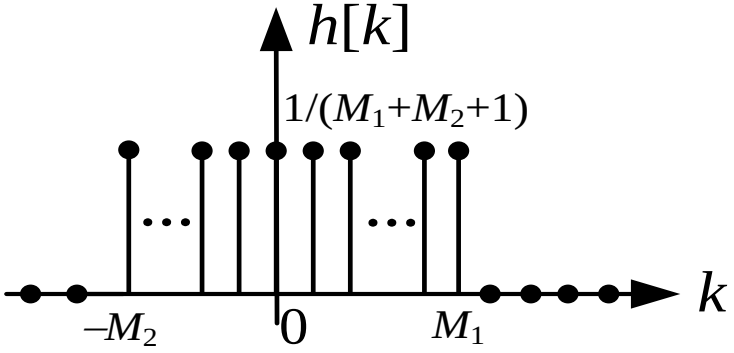
Therefore, the unit impulse response $h[k]$ of this system is given by:

$$h[k]=\frac{1}{M_1+M_2+1}\sum_{n=-M_1}^{M_2}\delta[k+n]$$

[E.g] Assess the causality and stability of the M_1+M_2+1 point moving average system.

Solution:

$$h[k]=\frac{1}{M_1+M_2+1}\sum_{n=-M_1}^{M_2}\delta[k+n]$$



The unit impulse response $h[k]$of this system is:

$$h[k]=\begin{cases}\dfrac{1}{M_1+M_2+1} & -M_2\le k\le M_1\\ 0 & \text{others}\end{cases}$$

[E.g] Assess the causality and stability of the M_1+M_2+1 point moving average system.

Solution:

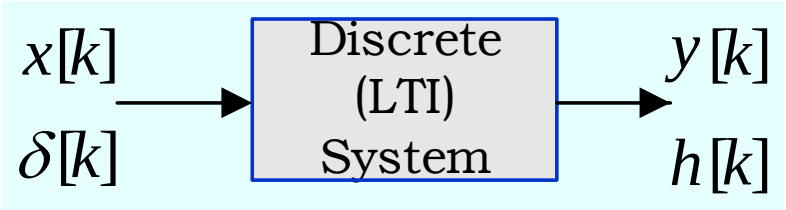
$$h[k]=\begin{cases}\dfrac{1}{M_1+M_2+1} & -M_2 \le k \le M_1 \\ 0 & \text{others}\end{cases}$$

Since this system is a discrete LTI system, when M_2 =0, it satisfies $h[k]=0$, $k<0$, indicating that the system is causal.

Since $\sum_{k=-\infty}^{\infty}|h[k]|=1$, Therefore, the system is stable.

Analysis of the Response of Discrete Linear Time-Invariant (LTI) Systems

The theoretical foundation of convolution lies in the time-domain representation of signals as a summation of delayed (or shifted) signals (δ-signals), thereby reducing the analysis of any signal acting on an LTI system to the analysis of δ-signals acting on the LTI system. This, in turn, introduces the time-domain characteristics of the system.



$$x[k] = \sum_{n=-\infty}^{\infty} x[n]\delta[k-n]$$

$$y[k] = \sum_{n=-\infty}^{\infty} x[n]h[k-n] = x[k] * h[k]$$

[E.g] Given the unit impulse response of a discrete-time LTI system $h[k]=(1/3)^k u[k]$, Input Sequence $x[k]=[1-(1/2)^k]u[k]$, try to find the zero state response of the system $y_{zs}[k]$。

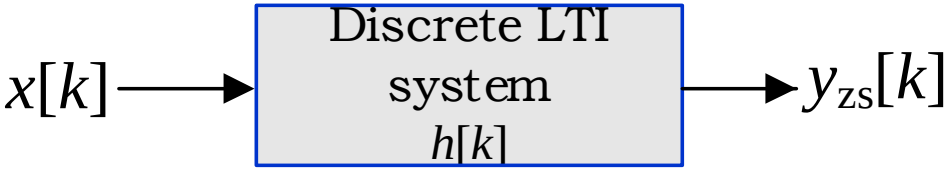
Solution: The zero state response of the discrete LTI system $y_{zs}[k]$ is:

$$y_{zs}[k] = x[k] * h[k] = \sum_{n=-\infty}^{\infty} x[n]h[k-n]$$

$$= \sum_{n=-\infty}^{\infty} [1-(1/2)^n]u[n](1/3)^{k-n}u[k-n] = \begin{cases} \sum_{n=0}^{k} [1-(1/2)^n](1/3)^{k-n}, & k \geq 0 \\ 0, & k < 0 \end{cases}$$

$$= [3/2 + 3/2(1/3)^k - 3(1/2)^k]u[k]$$

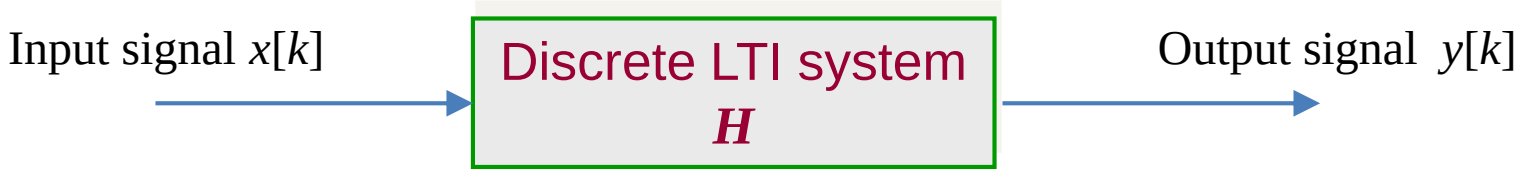
Analysis of the Response of Discrete Linear Time-Invariant (LTI) Systems



$$y_{zs}[k] = x[k] * h[k] = \sum_{n=-\infty}^{\infty} x[n]h[k-n]$$ (The convolution sum)

The zero state response of a discrete-time LTI system is the convolution of the input signal and the system's unit impulse response, which reveals the mechanism of signal system interaction in the time domain.

Engineering Applications of Sequential Convolution and



$x+H \rightarrow y$ Analysis

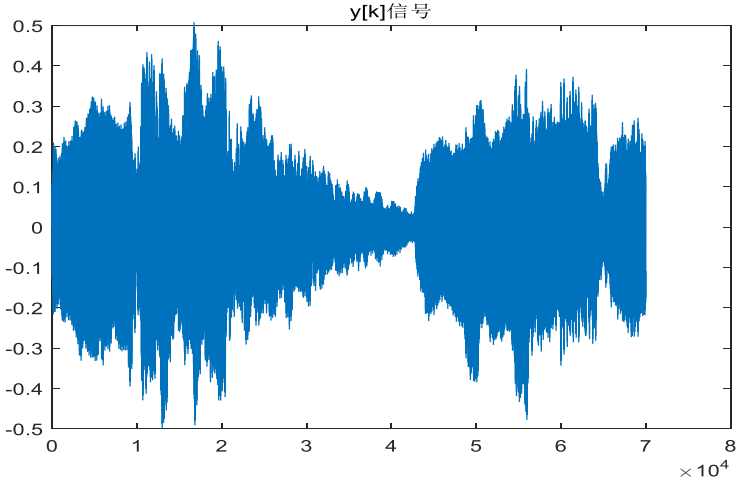
$y+H \rightarrow x$ Control

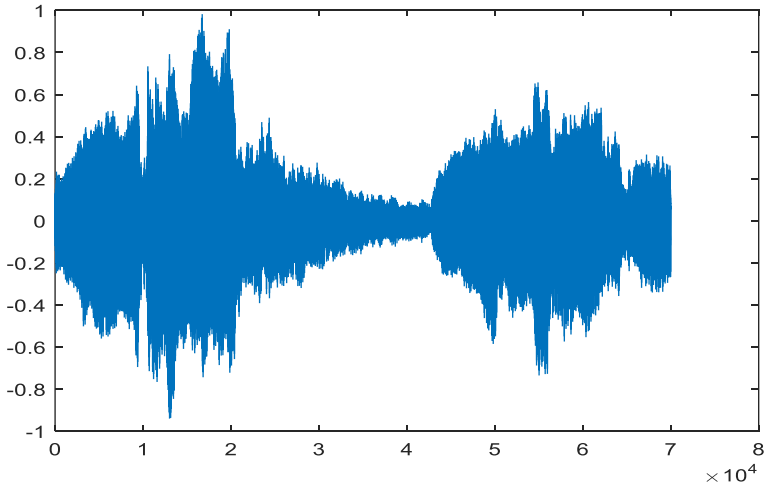
$x+y \rightarrow H$ Design

Engineering Applications of Sequential Convolution and

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Given the $y[k]$ waveform of the output audio signal of the system, which is obtained by passing the input music signal $x[k]$ through a low-pass filter system, the original music signal $x[k]$ is obtained by deconvolution.





MATLAB program code:

```
[y, fs] = wavread('E:\y.wav');                  %Read y[k]
time = (1:length(y)) / fs;figure;               %Calculate playback time
plot(time,y);                                   %Draw y[k] waveform
load('h2.mat');                                 %Unit impulse response
x = deconv(y, h);                               %Deconvolution
wavplay(x, fs);                                 %Play x[k]
time = (1:length(x)) / fs;                      %Calculate playback time
figure; plot(time,  x);                         %Draw x[k] waveform
wavwrite(x,fs,‘E:\x_new.wav’);                  %Save x[k] audio file
```

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.